

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The meeting focused on the development of proof of personhood and dual agent systems, with a significant portion dedicated to understanding how these concepts can be standardized and adopted by industry stakeholders. The discussion included the importance of separating information sharing and protection within dual agent architectures, the role of blockchain in adding time as a dimension to proofs, and the challenges and successes of standardization processes, particularly within ISO.

Key Discussion Points

1. **Dual Agent Systems:**
 - **Separation of Concerns:** Dual agents are necessary to handle conflicting constraints such as protecting information and sharing it.
 - **Trust Tasks:** Establishing trust through tasks that allow for gradual information sharing.
 - **Emergent Properties:** The ability to reflect on and connect information, creating a secret language between agents.
2. **Proof of Personhood:**
 - **Human Stack:** A model mirroring the IP stack, with UI and trust tasks at the top and credentials at the bottom.
 - **Blockchain for Time Measurement:** Using blockchain to add time as a dimension to discrete measurements, enhancing trust and verifiability.
 - **Knowledge Substrate:** The knowledge substrate is kept separate from cryptographic keys to ensure security.
3. **Standardization:**
 - **ISO Standards:** BGIN's role as a liaison to ISO TC307 allows for the submission of documents for publication.
 - **Types of Documents:** TR (Technical Report), IS (International Standard), and TS (Technical Specification).
 - **Adoption Challenges:** The importance of real-world implementation and industry buy-in, especially from major players like Apple and Google.
4. **Industry Adoption:**

- **IEEE 7012:** An example of a standard that took eight years to develop but is now moving towards adoption.
- **Smooth Entry Points:** Creating easy-to-implement initial features to encourage broader adoption.
- **Machine-Readable Personal Privacy:** The potential for standards like my terms to gain traction as AI consumes more data.

Action Items and Next Steps

1. **Continue Research and Development:**
 - Further explore the dual agent architecture and its emergent properties.
 - Develop a comprehensive model of the human stack and its interactions with blockchain and other technologies.
2. **Standardization Process:**
 - Prepare the personhood paper for submission to ISO as a TR (Technical Report).
 - Engage with industry stakeholders to ensure real-world applicability and adoption.
3. **Feedback and Collaboration:**
 - Distribute the meeting report to all participants for feedback.
 - Encourage ongoing collaboration within the working group to refine and expand the research.
4. **Future Meetings:**
 - Schedule regular meetings to discuss progress and address any challenges that arise.
 - Plan for a presentation at block 15 in October in DC to open up the paper to public comment.

Conclusion

The meeting concluded with a clear path forward for the development and standardization of proof of personhood and dual agent systems. The importance of separating concerns, ensuring real-world applicability, and engaging with industry stakeholders were emphasized. The next steps include continuing research, preparing documents for submission, and fostering collaboration within the working group.

Detailed Session Summary

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Appendix: Full Meeting Transcript (Anonymized)

28.4 --> 29.7: Hello, everyone.
29.7 --> 33.2: Let's get going on this session.
33.8 --> 35.9: I'm super excited for this session.
36.3 --> 42.7: I've been spending a lot of time researching into different proof of personhood protocols,
43.0 --> 47.7: different mechanisms, trust graphs, all of the above.
48.1 --> 52.0: So super excited to share some of those insights,
52.0 --> 59.2: hear from different protocols from the community that we have here today as well.
59.7 --> 65.0: I do need to join the Zoom call so I can share my screen.
65.3 --> 67.6: So I'll get that going now.
69.5 --> 76.6: But high level, I guess the concept that I want to put in to everyone's head,
76.6 --> 83.7: ahead of hearing the different solutions and the different ways of achieving proof of personhood,
84.0 --> 85.2: there are two questions.
85.7 --> 96.8: One is proof of personhood in terms of our access and control to systems.
97.1 --> 102.8: And then the other is accelerating faster than we can even realize.
103.2 --> 106.1: And that's proof of personhood as opposed to proof of robot.
106.1 --> 110.9: So how do we prove that we are a person making these changes,
112.8 --> 118.5: signing these things in systems versus it being an AI agent?
118.9 --> 127.6: And I think that personhood and decentralized identity, so DIDS, VCs,
128.1 --> 135.2: these have been something of the identity space and the cryptography space
135.2 --> 138.5: and like blockchain therefore.

139.3 --> 146.3: However, now that AI is really taking over and accelerating a whole bunch of things,
148.3 --> 152.1: these proof of personhood protocols and identity space is starting,
152.4 --> 154.8: like we all predicted it, now it's becoming mainstream.
155.5 --> 161.6: So it's time that we sort of get together and work out what the consolidation of this will look like.
162.6 --> 166.8: And the concept that I've been working on is three graphs, one overlap.
167.9 --> 169.7: And what are the three graphs?
170.6 --> 172.8: I know I'm that guy who always talks about graphs.
173.3 --> 174.1: It's crazy.
174.8 --> 176.0: But graphs are important.
176.4 --> 183.2: And when you put lots of graphs together, it starts to look like an overview of logic or consciousness,
183.4 --> 184.9: if you want to go that far.
185.5 --> 190.2: And then looking at the way that AIs create logic also goes in that direction.
190.2 --> 191.9: But yeah, three graphs, one overlap.
192.2 --> 193.2: So what's the first graph?
193.5 --> 195.0: The first graph is substrate.
196.0 --> 197.2: And this can be anything.
197.8 --> 210.1: This could be our gardening club book and all of the different flowers and techniques that we use for gardening.
210.3 --> 211.0: That is our substrate.
211.3 --> 214.3: That's our knowledge graph within this context, within this ecosystem.
215.2 --> 216.8: Now, what's the second graph?
217.0 --> 219.1: Well, it's promises and intent.
219.1 --> 224.9: And this gets into the intent-based economy, which is where we're going with a lot of these things,
225.0 --> 230.0: because especially when we delegate agency, intent is important.
230.6 --> 239.0: So the promise graph or the intent graph is the overlap that gives weight and action to the knowledge.
239.2 --> 245.6: So what we traverse, what we think about, how we flow through the system.
245.6 --> 249.9: And then what is the emergent graph from the overlap between these two?
250.5 --> 257.4: Well, over time, trust develops based on whether or not this knowledge is helpful,
258.0 --> 262.4: whether or not you follow through on contributing more information.
262.8 --> 266.3: So I came up with a new technique of how to water flowers.
266.6 --> 269.7: Adding that to the knowledge graph becomes a contribution.

270.7 --> 275.8: And what's interesting about this is these are the three systems that can sit separately,
276.0 --> 286.9: which means that the overlap or the coordinates of when knowledge meets promise equals trust is extremely unique.
287.4 --> 296.9: And I'll provide a visualization of this after we hear from a couple of the experts working on personhood protocols in the room,
297.9 --> 304.8: of what you can maybe envisage in the future where you don't, using zero knowledge proofs,
304.8 --> 308.3: you don't actually need to sort of prove any of these credentials.
308.5 --> 313.7: You prove that at one point in time, I went through a ceremony,
314.1 --> 318.5: whether or not that's a biometric ceremony or another type of ceremony of personhood,
318.7 --> 325.4: and I'm the only one who has gone on the trajectory from this point to that point.
325.4 --> 336.7: And that then eliminates any of the overburdensome sharing of information that we have in the existing system,
336.9 --> 339.5: where I have to show like the bartender problem,
339.6 --> 347.3: where I have to show the bouncer where I live when all they need to know is a range proof that I'm over 18 to get into the bar.
347.3 --> 364.4: So with that as context, I'd like to invite World to give a presentation on how they have supported with the biometric side and sharing a case study that's been based in Japan.
365.7 --> 366.7: So, yeah, go ahead.
373.1 --> 375.4: Hello, everyone.
375.6 --> 380.0: My name is Marcin Czarniecki and I'm the data protection officer for World Foundation.
381.9 --> 387.7: Firstly, I would like to thank Mitchell and Dr. Matsuo-san for having us.
388.4 --> 392.8: I'm here also with my colleagues, also from the World Foundation, Mark, Dom.
392.8 --> 404.2: We also have online Remco and Ian, as well as our partner company, Building for World, Just Tools for Humanity, Tomoa and Moon.
404.8 --> 412.3: Sorry for the long introduction, but, you know, feel free to speak to us after the conference.
412.3 --> 418.4: Please give me one second just to share my screen and I'll continue.
431.0 --> 431.7: Yeah.
432.6 --> 433.9: I think we're good.
435.2 --> 435.9: So.
441.4 --> 443.2: So let's start with the basics.
445.0 --> 452.1: I'll tell a little bit about why and what and then tell you more about how.
454.8 --> 467.4: So the 2025 IMPERVA reports on bots acknowledge that over half of the web traffic is now driven by bots.
469.2 --> 478.9: Exactly 51 percent and 37 percent of the traffic are bad

d bots, which is an increase from 2023 where it was 32 percent.

479.9 --> 482.6: And these numbers are probably going up.

483.4 --> 488.0: And those numbers relate to both Web 2.0 as well as Web 3.0.

488.2 --> 491.6: So this problem relates to all of us.

492.5 --> 496.4: Secondly, why are bots such a threat?

496.8 --> 501.2: Firstly, they can spread disinformation.

501.6 --> 504.6: They can affect digital discourse.

504.6 --> 514.7: They allow fraud on mass scale and basically they threaten basic human agency and the Internet as we know it.

516.4 --> 530.7: And one of the key things about those automated, I would say, agents is that they can be used by large groups and they can be used by small groups or individuals.

531.7 --> 542.2: Because technology nowadays allows us to create a lot of synthetic content at a low price and the price is going even down.

542.6 --> 550.2: The quality is going up and we cannot distinguish what is real, what is not.

550.2 --> 567.2: Even if we go online and interact with social media site or media site and read comments, we don't really know how to do that, how to distinguish what is synthetic, what is human.

567.2 --> 581.5: And that was one of the things that was on the mind of the creators, the founders of the project, which was Sam Altman and Alex Blania.

582.3 --> 595.6: And many years ago they started thinking that the test that we have will at some point break and future came quicker than intended.

595.6 --> 604.5: And nowadays all the CAPTCHA test can be solved by AI easily or the email or phone verification can be duplicated at scale.

606.9 --> 620.2: And the founders came to the conclusion that Internet needs a foundational building block, a proof of human, which is anonymous and private.

625.5 --> 630.0: And they thought that there should be two primitives around that.

630.4 --> 634.8: Firstly, humanness, which is checking whether somebody is human.

635.6 --> 642.2: And secondly, uniqueness, i.e. whether somebody is a unique human without saying which human.

642.5 --> 645.6: So we can see here, is this the first time I'm seeing you?

649.3 --> 651.0: And this is very important.

651.2 --> 654.8: These are very powerful, I would say.

654.9 --> 657.9: This is very powerful to be able to assert in that.

658.0 --> 669.4: And you can see in that graph how different is proof of human from digital authentication, which answers the question, are you who you say you are?

669.5 --> 673.0: Or digital identity, which just say who you are.

673.0 --> 675.2: This is much more basic.

677.2 --> 682.8: And when they were discussing this, they thought that in order to perform such a proof of human,

683.4 --> 688.4: their view that there needs to be a link to something real, something in the physical world.

689.0 --> 694.4: And what they came up with is that this should be biometrics.

694.4 --> 699.8: Because biometrics, especially iris recognition, can work on a global scale.

700.0 --> 706.2: It can be scaled to one billion people and is very accurate.

709.5 --> 713.4: And that's where the orb was born.

714.2 --> 718.8: Firstly, they thought, okay, maybe we can use any tooling of the shelf.

718.8 --> 724.0: But it turned out that none of them met the requirements.

725.1 --> 734.1: So for a couple of years, engineers built an orb, which is that super fancy, sophisticated camera at its core.

734.2 --> 736.2: But it also has some additional features.

737.2 --> 758.3: It has an NVIDIA Jetson within its running AI models responsible for anti-spoofing, for calculation of iris codes, which I will tell you in a second.

759.3 --> 768.5: Some of the interesting things about this piece of equipment is that all the calculation is done on the orb.

770.4 --> 775.2: And basically, it's designed to work in an untrusted environment.

775.5 --> 781.1: So all the security measures are very high, especially to prevent any intrusions.

782.3 --> 786.6: It's auditable. Every orb can be audited separately.

786.6 --> 790.8: And what's more, core components have been outsourced.

791.5 --> 802.0: So this is the tool how to make sure that we can create proof of personhood.

802.6 --> 809.8: And let me now tell you how it works, which is probably the most interesting thing.

810.6 --> 817.2: By the way, apart from this orb, which is just for demonstration, there is also a real working orb in the other room.

817.6 --> 827.1: So feel free to go there, watch it, check it out, and we'll be happy to answer any additional questions about it.

827.8 --> 838.0: So how does the ceremony, as Mitchell said, which is a very good framing of this, how does it work?

838.0 --> 844.4: Firstly, you need to download an app. It's called World App. It's a propriety app for WorldProject.

844.9 --> 850.7: It allows to store World ID. It allows to generate proofs related to proof of human.

852.3 --> 856.2: But it's also much more. It's a non-custodial wallet.

856.9 --> 858.9: It also has a chat function.

858.9 --> 870.4: It also has a whole store of so-called mini apps, which are added by developers with various functionalities.

871.7 --> 880.1: So once you find an orb, and it's possible to do that through WorldApp, you start with the verification process.

880.1 --> 892.8: So firstly, when you first open your WorldApp, your World ID is created as a public-private key pair that is created.

893.7 --> 905.4: Then, using your QR code on the app, the app is connecting with the orb to establish an air-gapped communication.

908.3 --> 914.5: And the next step that you can see, maybe I can have a pointer, is taking a photo.

915.3 --> 917.6: The orb is taking a photo of iris and a face.

918.2 --> 925.9: And based on neural networks running on the orb, it checks whether you're a real human.

926.4 --> 933.7: Whether that's not a photo, whether that's not somebody wearing eye lenses for spoofing.

934.7 --> 940.9: When that humanity is confirmed, then there is another face.

940.9 --> 943.3: Of course, it's seamless for a user.

944.6 --> 952.0: But the other face is confirming the uniqueness, whether the person hasn't been in front of the orb yet.

952.7 --> 959.5: And how it's done, the iris photo is turned into a so-called iris code.

959.5 --> 963.7: Iris code is just an abstraction of the photo.

963.9 --> 970.9: It's a binary code with 212,000 digits.

971.7 --> 983.8: And that iris code is then encrypted using our anonymous multiparty computation technology into AMPC fragments.

984.6 --> 990.6: And this is very unique, because each of those fragments doesn't mean anything.

991.2 --> 993.0: And it's useless on its own.

993.2 --> 1000.3: And what happens then is that the encryption protocol checks whether the person has been before.

1001.4 --> 1014.2: And the homomorphic properties of that system allow the verification to be done without decrypting the fragments.

1014.6 --> 1018.6: And the calculation is done on the fragments separately.

1020.6 --> 1025.1: So what's more, the protocol is designed in a way that the only outcome is yes or no.

1026.1 --> 1034.2: And if the answer is no, meaning that such a person has not been in front of the orb, the world ID is verified.

1034.4 --> 1038.1: Which means that going forward, the person can prove that they're human.

1038.3 --> 1040.1: I'll tell you how it's done in a second.

1040.6 --> 1043.7: But there is one important thing happening in the background.

1045.5 --> 1053.6: That all the photos, iris codes are encrypted, signed, and sent to the user's phone.

1053.9 --> 1058.0: So that all the images are only on the user's phone.

1058.3 --> 1059.6: And they are deleted from the orb.

1061.2 --> 1064.1: And that's how the entire process looks like.

1065.1 --> 1077.1: Now, one important word about our AMPC, the multi-party computation technology, is that...

1077.1 --> 1084.5: I think this is a cryptographic breakthrough that allowed this project to be very private and very secure.

1085.0 --> 1088.3: And also to be decentralized.

1088.3 --> 1094.8: Because none of the World Foundation, Tools for Humanity, is holding those AMPC fragments.

1095.5 --> 1099.8: This is held by trusted third parties, being well-known universities.

1100.4 --> 1105.7: Currently these are Berkeley RDI, being part of the University of Berkeley.

1106.5 --> 1110.4: The University of Erlangen-Nuremberg in Germany.

1110.7 --> 1114.9: And the Korea Advanced Institute of Science and Technology.

1114.9 --> 1125.6: So these are the nodes having the AMPC, being part of the AMPC network.

1126.2 --> 1131.9: And what happens next is that you can have your...

1131.9 --> 1137.9: Now you have your World ID, which is a privacy-preserving, permissionless protocol.

1138.0 --> 1142.1: And it allows you to create zero-knowledge proofs to prove that you're a human.

1143.8 --> 1152.5: Moreover, we introduced a feature called World ID Credentials, which allows also to create proofs about other topics.

1153.0 --> 1163.4: Especially information scanned from NFC-enabled chips that can be stored on your phone.

1163.4 --> 1172.0: And based on that information, zero-knowledge proofs can be performed about specific age requirements or citizenship.

1172.7 --> 1178.2: And those credentials can be developed.

1178.6 --> 1182.0: There is an SDK for developers to create more credentials.

1186.5 --> 1192.4: As I said, all those World ID proofs are done using zero-knowledge proofs.

1192.4 --> 1198.5: I don't want to tell more about it, because probably the audience here knows about zero-knowledge proofs more than I do.

1200.1 --> 1206.0: But I'd like to just quickly move to use cases, because that's an important thing for the discussion.

1206.7 --> 1213.9: The first one is our cooperation with Razer, which is a company creating an ecosystem for gamers.

1213.9 --> 1220.9: And they have their Razer ID, which is a single sign-on and an account for the user,

1221.1 --> 1231.5: which can verify that their Razer ID is held by a human with a human badge.

1232.0 --> 1236.3: Which changes the experience of playing, for example,

1236.9 --> 1240.8: that people know that on the other side of the computer there is a real human.

1240.8 --> 1244.6: But it's important, in any case, not saying what is the human.

1244.7 --> 1246.0: That's the beauty of the World ID.
1246.1 --> 1250.3: It doesn't in any way say who is the human, just that it's a unique human.
1250.8 --> 1254.4: The second use case, very important here in Japan, is Tinder,
1255.0 --> 1264.0: where our technology has been implemented firstly to confirm that users are above 18.
1264.5 --> 1268.3: And again, this is done using zero-knowledge proofs,
1268.3 --> 1279.0: meaning that such information as the date of birth or the exact age are not revealed.
1279.2 --> 1285.4: Only the information that you're above 18 or other age that somebody might think about.
1285.6 --> 1290.3: The second one is the same, a possibility to verify that one is a human
1290.3 --> 1294.1: and placing a human badge on the account,
1294.1 --> 1299.3: knowing that the person we're interacting with is a real person, not a bot.
1299.5 --> 1303.8: Which, in the case of Tinder, is probably a very important thing.
1305.1 --> 1313.3: And I'll finish with just a slide showing what are some of the applications that World ID can help with.
1313.4 --> 1321.7: And I believe that there are not dozens, but hundreds or thousands of those that can be done.
1321.7 --> 1324.9: There is a lot, especially in the web-free space,
1325.2 --> 1333.6: but I think that this is a technology that is needed for all of the internets,
1333.7 --> 1337.0: whether web 2 or web 3.
1338.0 --> 1339.8: And that's basically it.
1340.2 --> 1343.6: Before I hand over to Mitchell, I'll just ask anyone from our team
1343.6 --> 1345.3: whether anyone has anything to add.
1358.0 --> 1359.5: Thank you for sharing.
1360.1 --> 1362.9: Quickly, if there's anyone in the audience online,
1363.0 --> 1367.2: feel free to either put your hand up or put a note in the chat.
1367.4 --> 1370.5: I'm in the channel, so I'll be able to read the question out.
1371.5 --> 1379.6: So, as I explained earlier, I see World as a knowledge or a knowledge substrate
1379.6 --> 1383.1: for proof of personhood in terms of the overlap.
1383.7 --> 1393.0: So, if biometric proofs are important, and as we identified in dating, it's probably important,
1394.2 --> 1396.6: perhaps we should use biometrics.
1396.6 --> 1401.2: However, there are lots of different types of proof of personhood,
1401.7 --> 1405.8: and I would suggest over time it becomes the unique overlap of these

1405.8 --> 1413.2: that we as an individual compose in order to get access to systems or create relationships.

1414.5 --> 1419.6: And that's in line with another project called the First Person Project,

1420.0 --> 1424.6: and there are two types of credentials that have been identified in the First Person Project.

1424.9 --> 1427.7: The first is a personhood credential.

1428.4 --> 1433.1: So, I would see World ID as a personhood credential.

1433.4 --> 1440.1: So, a personhood credential is a top-down, an ecosystem is providing a credential to an individual

1440.1 --> 1443.5: to recognize that they're a member in their group.

1443.9 --> 1450.7: So, Begin is an ecosystem, and we would provide a credential to the people within our ecosystem

1450.7 --> 1455.3: based on the governance that we have at Begin and what we think is a personhood credential.

1455.8 --> 1462.0: And perhaps that requires biometrics for some, perhaps it requires social proofs for other,

1462.5 --> 1469.8: or knowledge proofs or human-centric verification around artifacts or stamps,

1469.9 --> 1475.7: which is another alternative of personhood that has been discussed and explored in our research

1475.7 --> 1480.4: of the IKP Working Group in particular.

1481.1 --> 1489.8: So, the other type of credential is once you are proven to be a person in an ecosystem or network or in a graph,

1490.4 --> 1494.7: you have the ability to create bilateral relationships with people within that graph.

1495.2 --> 1501.4: And this is called a verifiable relationship credential, and it's specifically between two nodes.

1501.4 --> 1506.4: And this type of credential is new, and it's what I'm most excited about

1506.4 --> 1514.6: when it comes to the progression of proof of personhood in digital networks or cyberspace,

1515.0 --> 1522.1: because it is this genuine ceremonial connection that you have to an individual

1522.1 --> 1525.6: that over time generates the trust graph.

1526.3 --> 1531.1: And in order to enter the system, you need a personhood credential,

1531.1 --> 1536.2: but then in order to enhance yourself or in order to be a part of the network

1536.2 --> 1539.2: and create connections with individuals throughout this network,

1539.4 --> 1547.4: you have these key ceremonies or these verifiable relationship credentials or VCs attached to DIDS

1548.1 --> 1554.1: that then over time aggregate a social graph or a social proof

1554.1 --> 1559.4: that then can overlay with the knowledge graph of the network or the ecosystem.

1560.0 --> 1569.1: So First Person Project is the first, I don't want to call it an experiment,
1569.6 --> 1574.7: but the first, I guess, customer of First Person Project is the Linux core kernel.
1575.9 --> 1578.9: And this is like a highly secure system.
1579.3 --> 1583.5: The kernel is one of the most important technologies that we have in open source.
1584.5 --> 1589.7: And we're trying to use the First Person Project like dual credential system
1590.5 --> 1595.4: for supporting those maintainers of the Linux kernel
1596.5 --> 1604.6: and the maintainers access and ability to sort of merge, push, add code to the kernel project.
1605.1 --> 1608.9: So that will be happening over the next six months to a year.
1609.9 --> 1613.9: And I will report back to the group on how that all goes
1613.9 --> 1621.1: and whether or not that experiment can then be used to sort of a greater community
1621.1 --> 1622.9: of having these sort of dual credentials.
1624.7 --> 1627.8: I'm going to pause there because I don't want to just lecture at everyone.
1628.0 --> 1632.0: I want to make sure that the conversation is involved in a workshop.
1632.0 --> 1638.4: So has anyone got any comments or ideas on that or the overlap with the world biometric?
1642.0 --> 1643.0: Pause for a second.
1650.7 --> 1651.4: Okay.
1653.0 --> 1654.1: I'm going to share my screen now.
1663.1 --> 1665.1: This is just the agenda for today.
1665.7 --> 1667.4: Here's the First Person Project.
1669.0 --> 1670.9: It's the white paper here.
1672.6 --> 1673.5: I think.
1674.1 --> 1675.3: Oh, my screen is.
1677.1 --> 1684.1: And here is the diagram that explains ecosystem
1685.3 --> 1689.8: then invites a person, they receive credential
1689.8 --> 1692.9: and then you get this bilateral system that rotates through.
1695.6 --> 1697.9: And what's interesting about this is it creates
1699.3 --> 1705.0: the person who credential is the node or the vertex within the trust graph
1705.0 --> 1708.7: and then the verifiable relationship credentials are the lines
1708.7 --> 1711.7: between the people within this trust graph.
1714.2 --> 1716.6: So something that I've been working on,
1716.6 --> 1720.1: which is trying to be an expression of a knowledge graph.
1720.8 --> 1724.1: So when I say knowledge graph or substrate,

1724.7 --> 1728.8: knowledge graph in this case is all of the research that I've been doing
1730.9 --> 1732.3: on agent privacy.
1733.1 --> 1734.9: So you could imagine this here.
1736.8 --> 1741.6: It's something that you as an observer is looking at.
1741.6 --> 1748.3: And me, I was the one who defined the paths through this system.
1749.0 --> 1752.7: So I picked the edges.
1753.5 --> 1757.9: I've written all of the content which created this knowledge graph.
1758.5 --> 1762.7: And when an AI goes through this knowledge graph or through this system,
1763.7 --> 1766.4: it has a fairly defined path.
1767.3 --> 1770.5: But if a human was to observe this knowledge graph
1770.5 --> 1772.0: and then choose its own path,
1773.2 --> 1778.9: then we could have what I call a constellation of observing,
1779.7 --> 1785.8: which is a very unique thing that you could do with someone else's knowledge.
1786.7 --> 1793.0: So if I was to go and sort of learn about or have my AI
1793.0 --> 1794.8: learn about someone else's knowledge
1794.8 --> 1798.8: and then track my own path through that knowledge substrate,
1800.7 --> 1803.6: and then provide a proof of this constellation,
1804.4 --> 1805.5: download it as an MD,
1807.0 --> 1809.9: and then do some sort of zero-knowledge magic
1809.9 --> 1814.9: and put it behind a credential, which is either a DID or a VC,
1815.6 --> 1818.7: then I could prove that I observed this knowledge substrate
1819.5 --> 1824.8: and then I'm able to involve myself with this community in a more profound way
1824.8 --> 1828.9: because my trajectory here uses all of these nodes
1829.8 --> 1832.2: and then the overlap with the other person
1832.2 --> 1835.3: that I'm trying to create a relationship credential with
1835.3 --> 1839.7: would be able to verify that those nodes do exist in the knowledge graph
1839.7 --> 1842.3: and I've observed those nodes as well myself.
1842.8 --> 1845.7: But I noticed that you took a very different path to what I did.
1846.0 --> 1849.5: So where is our overlap in interest within this knowledge substrate?
1849.7 --> 1851.4: So do we care about ZKPs?
1853.2 --> 1856.1: Which has like a very different path.
1856.1 --> 1863.3: So one thing that I'd like to show here is I typed ZK

P

1863.3 --> 1866.3: and this is very random.
1866.7 --> 1871.5: There's like a lot of entropy here in terms of how ZK
P pathed
1871.5 --> 1874.9: and if I was to highlight one here
1874.9 --> 1878.3: and then add it to the constellation
1880.4 --> 1883.3: and then highlight another one.
1883.3 --> 1886.3: Wait, sorry. Let's clear this out.
1889.3 --> 1894.1: I hope the group doesn't mind me sort of using you guys
1894.1 --> 1896.9: as a bit of an experiment to share.
1901.5 --> 1906.8: But let's start a waypoint and go through like a little
experiment together
1908.2 --> 1910.8: and maybe I'll get some involvement from the room.
1912.0 --> 1913.8: So we'll start here
1916.1 --> 1918.5: and we'll start a waypoint
1920.6 --> 1926.3: and someone in the room pick one of these that they're
most interested in.
1926.6 --> 1928.3: One of these lines for us to follow.
1932.6 --> 1934.1: Do I have to point to someone?
1937.5 --> 1938.9: Which book of promises?
1939.4 --> 1940.7: Yeah, promise theory is interesting.
1941.4 --> 1942.7: So we'll add that to the path.
1946.6 --> 1948.2: Next. Eric?
1951.7 --> 1953.9: Oh, how are we going to get to Zcash from here?
1954.0 --> 1954.3: Okay.
1954.8 --> 1958.1: Well, I could help you because I created this knowledge
graph, right?
1958.8 --> 1959.9: That's part of the point.
1959.9 --> 1967.7: Yeah, so, well, I would say the first person spellbook.
1969.5 --> 1970.7: That's in the path.
1971.4 --> 1972.6: We'll go back that way
1974.9 --> 1977.0: and we'll do the Zcash shield.
1982.5 --> 1984.1: And we'll add that to the path.
1985.3 --> 1986.6: Okay, from the Zcash shield,
1986.6 --> 1991.0: who wants to add another node to our journey through
my knowledge substrate?
1992.8 --> 1993.6: Go for it.
1997.7 --> 1998.9: Triangle geometry.
1999.7 --> 2000.8: The tetrahedrons.
2001.7 --> 2003.5: Very interested in that shape.
2004.7 --> 2011.4: So the spell that's attached to that is also that memory
can never touch action.
2012.0 --> 2013.1: Memory is infinite.
2013.5 --> 2015.8: Action is discrete measurements in the world.
2015.8 --> 2019.0: That's what that triangle of geometry means.

2019.9 --> 2020.7: Next one.
2021.3 --> 2022.3: Come on, this is a fun activity.
2022.4 --> 2024.4: Nikita, tell me which one.
2028.2 --> 2029.0: Golden ratio.
2029.6 --> 2030.7: We like the golden ratio.
2030.9 --> 2031.7: The Nodulus shell.
2035.3 --> 2040.1: And then, oh, I'm going to pick dragon emergence because we're at the dragon gate.
2040.4 --> 2041.3: We should do that.
2044.1 --> 2045.6: And then progressive trust.
2048.0 --> 2052.3: And look, we found verifiable relationship credential, which I was talking about before.
2053.4 --> 2054.7: So we'll add that to the path.
2055.8 --> 2060.0: And then, oh look, we found three-layer identity, which I was talking about as well.
2061.5 --> 2062.9: So we'll add that to the path.
2063.4 --> 2064.4: Okay, we'll close the portal.
2069.0 --> 2072.9: Oh, I'm looking at the big screen and I'm clicking on the Zoom thing.
2073.7 --> 2087.9: So now, in this step, I'm creating a unique string to define almost like the machine personal description of this path through Mitchell's knowledge substrate.
2089.8 --> 2091.8: We could do a reflect quality.
2092.8 --> 2094.7: We'll just call it again.
2102.7 --> 2105.2: And save that as a particular constellation.
2108.8 --> 2110.8: So this does two things.
2111.8 --> 2130.8: Firstly, it gives us this very unique path that we as a group went through a knowledge substrate together, which if I download it, I could potentially use it as an overlap with the knowledge substrate.
2130.8 --> 2143.6: If maybe I crossed within this experience, I could write a promise to say that I'm going to share this with everyone who participated in this room.
2144.2 --> 2156.2: And then we will do a sort of series of bilateral key ceremonies to appreciate that this constellation through Mitchell's knowledge graph is something that we went through.
2157.0 --> 2165.5: And then finally, I can download this and share it with an AI agent, which then defines the path through the knowledge substrate.
2166.8 --> 2173.9: So the AI agent, for example, and I'm going to bring this to world so then we can get a little bit discourse here.
2174.6 --> 2178.9: If within this knowledge substrate there was I have a world ID.
2180.3 --> 2185.8: And I need to click that in order to enter a particular sort of reason.
2186.2 --> 2196.3: When I create that overlap of my knowledge substrate, I'll click world ID and make sure that the agent follows that path in or

der to do the verification.

2197.5 --> 2200.0: So, no, this is like a little bit of.

2201.2 --> 2205.2: I'm still in the research stage here. I'm happy to share it with you all.

2205.5 --> 2212.7: It's a little bit raw, but I'm hoping that everyone will have their own constellation through my knowledge substrate.

2212.7 --> 2223.5: And then over time, there'll be an aggregation of these constellations, which will help improve the AI model that learns from this because it creates an overlap.

2223.6 --> 2226.3: So when someone else discovers something new.

2226.9 --> 2235.4: While observing my information, then I can add that if I think it's important to the progression of the research.

2240.9 --> 2241.4: Yep.

2249.9 --> 2254.7: So really the point of this is to reduce redundancy and knowledge sharing.

2255.1 --> 2263.6: And so when you have these calls, you don't have to go over everything all the time and people can just catch up very quickly based on what they already know.

2263.6 --> 2268.1: And then just catch up and learn the things that they missed out on.

2269.2 --> 2272.5: Yeah. So that is definitely one of the outcomes.

2273.0 --> 2276.8: And now imagine if you had your own version of this second brain.

2278.3 --> 2289.2: And what we could do is create based on sort of the amount of information that like based on our relationship, how much information we're comfortable sharing with each other.

2289.2 --> 2297.9: We could then create an overlap between where you get your knowledge substrate and use my edges and domains.

2298.2 --> 2301.9: And then we insert that into this graph.

2302.0 --> 2310.7: And then it creates an overlap, which then allows for discovery without disclosure of all of the information.

2310.7 --> 2320.9: Which in my opinion is kind of like the golden snitch of personhood is discovery with anonymity or obfuscation or zero knowledge.

2321.4 --> 2324.2: Discovery with zero knowledge is the best way to put it.

2326.3 --> 2336.2: So if everyone in this room had their own version of this and created an overlap, then we could pinpoint the coordinates where everyone's overlap is the strongest.

2336.2 --> 2342.4: And then hone in on that idea and expand from there is kind of the concept.

2342.6 --> 2347.2: And, yeah, you've picked up on removing these redundancies.

2347.5 --> 2359.2: But also when we delegate authority to AI, hopefully this will help the pathing that that AI takes based on the relationships that we have with each other.

2363.3 --> 2367.2: Cool. And I have a friend who's very interested in th

is, so I'll connect you later.

2367.7 --> 2373.6: Yeah, I am. And I built this fairly recently.

2373.9 --> 2384.2: So I'm going to open it up and find a way that we can put a lot of the begin work into this style of visualization.

2385.7 --> 2388.2: So then, yeah, we can continue the experiment.

2398.4 --> 2405.5: So the other thing that I'll just quickly share here, this is the AI agent version.

2405.5 --> 2410.6: So what you can do is you choose a persona.

2411.0 --> 2412.2: So we'll choose the mage.

2412.8 --> 2417.9: And then you also get a series of skills, which are the dot skills files.

2419.8 --> 2423.6: And then that creates its own graph here.

2424.3 --> 2427.9: You could then download that and then overlap it here.

2427.9 --> 2432.2: So you could imagine this is the whole knowledge substrate.

2432.6 --> 2436.4: And this is your personal specific AI agent's version of that.

2438.6 --> 2445.6: OK, I'm going to stop presenting the screen unless you guys think it's pretty.

2448.1 --> 2450.9: We can leave it here because it is kind of pretty.

2457.9 --> 2460.7: The universe of agent privacy.

2467.9 --> 2469.1: All right.

2473.6 --> 2474.9: Let me check.

2479.2 --> 2485.2: Yeah. Is there anyone online who would like to jump off mic?

2488.2 --> 2489.4: Make comments.

2499.6 --> 2502.4: Should I call Ian out or Remco?

2503.7 --> 2504.4: Both?

2507.3 --> 2509.8: Remco, I've wanted to speak to you about this.

2510.5 --> 2513.1: What is your initial view?

2517.9 --> 2519.3: Maybe he's not present.

2521.5 --> 2523.5: Is my microphone working?

2524.1 --> 2525.2: Yeah, we can hear you.

2525.5 --> 2525.8: Oh, cool.

2529.1 --> 2531.6: I generally like knowledge graphs.

2532.5 --> 2540.1: They were a big thing in, I think it was even pre-Web 2.0.

2540.6 --> 2540.9: Yeah.

2542.1 --> 2549.0: And you had these giant ontologies and triplet stores and all that cool stuff.

2551.4 --> 2558.8: And I think from there on this field sort of evolved into formal reasoning

2558.8 --> 2563.1: where Lean is now leading the pack and into machine learning

2563.1 --> 2568.5: which went into the complete fuzzy direction where Transformers are now leading the pack.

2573.1 --> 2574.0: Interesting.

2575.1 --> 2580.7: Yeah, but as a foundation for a modern stack,
2584.3 --> 2589.0: I guess GraphQL is kind of the modern interpretation
of that
2589.0 --> 2592.4: and there's also the Web3 version of that, right?
2593.0 --> 2597.0: So it's being used to model backend data structures
2597.0 --> 2601.6: and allow for efficient consumption of the data.
2607.5 --> 2608.5: Yeah.
2609.0 --> 2613.3: And I guess the question that I would pose to you is
2614.0 --> 2619.0: do you think that rather than the knowledge graph
2619.0 --> 2623.6: or being only observable by the AI,
2624.6 --> 2628.8: if we can create a system where I can expose my knowl
edge graph
2628.8 --> 2633.8: and create a signal in the world and someone can obse
rve that
2633.8 --> 2637.8: and then create their own path through that,
2638.9 --> 2643.5: is that, in your opinion, a uniquely human thing?
2643.6 --> 2650.4: Or do you think an AI over time will be able to get a
way from its efficiency
2650.4 --> 2657.5: in pathing knowledge in order to spoof the...
2657.5 --> 2657.7: Right.
2658.0 --> 2663.6: I am fairly confident that any observable human behav
ior,
2664.0 --> 2665.6: at least in the digital space,
2665.9 --> 2673.7: an AI can perfectly replicate over time if we're not
there already.
2674.7 --> 2679.4: But what is the concrete problem you're trying to sol
ve in that example?
2681.3 --> 2690.4: The concrete problem is building the overlap between
knowledge
2690.4 --> 2696.1: and then human promises within that knowledge.
2696.7 --> 2700.4: So if someone observed the knowledge substrate, went
through their own path
2700.4 --> 2705.5: and then added two emergent qualities, one being refl
ect,
2705.5 --> 2710.2: so their own reflection of that, and then the other b
eing connect,
2710.5 --> 2714.8: so they choose to make their own connection on the kn
owledge graph.
2715.1 --> 2716.6: So here you connect.
2716.7 --> 2717.0: Yeah.
2717.3 --> 2723.1: So this person, what are they trying to achieve that
day?
2724.5 --> 2731.9: They're trying to achieve increased information shari
ng
2731.9 --> 2735.4: between each other in a progressive trust way.
2736.7 --> 2738.4: Like what kind of information?
2739.3 --> 2744.7: Either research or potentially personally identifiabl

e information

2744.7 --> 2748.7: or mostly in the research.

2751.3 --> 2754.7: So they have a common research project that they're engaging in?

2755.9 --> 2760.5: Maybe not even a common research project, but cross-discipline as well.

2760.5 --> 2761.1: Gotcha.

2763.6 --> 2770.2: So being able to write your own interpretation onto someone else's graph

2770.2 --> 2776.6: using these reflect and connect qualities is fairly unique.

2779.5 --> 2785.6: I understand and I appreciate your comment that an AI will always be able

2785.6 --> 2791.5: to mirror or reflect or recreate, reconstruct a human's behavior

2791.5 --> 2796.7: if it has enough information of that human to reconstruct them.

2797.4 --> 2801.8: However, I guess one thing that I'm working on is ensuring

2801.8 --> 2805.8: that a single AI can't collapse into all of that information,

2806.0 --> 2808.7: so we need duality of the information.

2809.1 --> 2814.9: And one of those areas is the knowledge and the other, in my opinion,

2814.9 --> 2819.6: is the keys or the cryptography, which is where, yeah, biometrics.

2819.7 --> 2825.1: Right, but it's a different question whether an AI can pretend

2825.1 --> 2830.0: to be a real-life human being versus whether it can pretend

2830.0 --> 2839.7: to be a specific human being, because being a credible imposter

2839.7 --> 2843.2: is a lot harder than being just a credible individual, right?

2843.2 --> 2844.2: Yeah.

2846.1 --> 2846.8: Interesting.

2851.2 --> 2855.7: And to me, those are two very different problems.

2855.9 --> 2860.8: I think for proof of human, we're primarily focused with how do we make sure

2860.8 --> 2867.1: that it is a human being, and then from that, you build on more identity.

2868.1 --> 2873.0: Then you get into, like, is this specific individual

2874.2 --> 2876.8: would be the extreme version of it.

2878.1 --> 2882.7: A way more relaxed version of that would be is this an individual

2884.0 --> 2888.1: of at least 18 years of age, right, which is a very large category.

2888.1 --> 2897.4: In fact, within the world, we tend to stay very far
2897.4 --> 2899.2: from these specific categories.
2900.7 --> 2905.1: Like, the core concept in the system is anonymity,
2905.6 --> 2912.5: so we wouldn't want to be able to identify humans
2912.5 --> 2916.7: to within categories smaller than, let's say, a milli
on people,
2916.7 --> 2919.7: because then you're de-anonymizing too much.
2921.6 --> 2929.0: Of course, it is useful to create tools that can do t
his for other reasons.
2929.3 --> 2936.4: Like, sometimes you do want to give a very narrow pre
sentation
2936.4 --> 2940.4: of yourself to someone, and then it's cool to have to
ols
2940.4 --> 2942.7: that allow you to do that, but still be restricted.
2942.7 --> 2947.7: Like, you would be able to say that, well, I was born
on a Tuesday
2947.7 --> 2957.0: in this village without necessarily revealing who exa
ctly you are
2957.0 --> 2961.4: or where you currently reside or, I don't know,
2963.5 --> 2968.5: what your last year's taxable income was or whatever
it is.
2968.5 --> 2972.8: There's a lot of stuff that you still want to hide ab
out yourself
2972.8 --> 2976.1: while you might want to reveal something very specifi
c about yourself
2976.1 --> 2979.3: for all sorts of purposes, like, let's say, local pol
itics.
2982.6 --> 2988.8: And for that, I think we need much better zero-knowle
dge identity tooling
2989.5 --> 2996.3: where you as an individual have all the evidence abou
t yourself,
2997.6 --> 2998.3: but...
2998.3 --> 2999.6: Need help composing it.
3002.1 --> 3008.7: Yes, composing it, but also very selectively disclosi
ng it
3008.7 --> 3012.3: or rather than disclosing it in the traditional sense,
3013.4 --> 3014.8: ZK disclosing it.
3015.0 --> 3019.9: So being able to use all the evidence that you have a
bout yourself
3019.9 --> 3023.5: in the clear text, like, let's say, your digital birt
h certificate
3023.5 --> 3029.9: and whatnot, and then be able to prove to a third par
ty
3031.2 --> 3034.7: only the bare minimum that needs to be done
3035.5 --> 3039.0: for whatever the two of you are together trying to ac
hieve
3039.0 --> 3044.3: and also ideally in ways that this third party cannot

collect the data

3044.3 --> 3049.4: or pass it on to other parties and so on.

3054.2 --> 3054.8: Yeah.

3055.4 --> 3059.3: I think that I catch where you're coming from as well,

3059.4 --> 3066.1: and I guess this potential, this tool could also be potentially good

3066.1 --> 3070.8: for that composing of what the path is and making that unique

3070.8 --> 3074.6: and then wrapping that in a zero-knowledge proof

3074.6 --> 3077.5: because you just need to prove that you went on a path

3077.5 --> 3082.4: through the information that is related to the outcome

3082.4 --> 3083.3: that you're looking for.

3083.5 --> 3085.6: So, for example, to be proven over 18.

3086.8 --> 3092.3: Yeah, and to some extent we're seeing the early developments

3092.3 --> 3096.1: of that within our stack.

3097.9 --> 3104.4: With the orb, you receive a piece of evidence in the clear

3104.4 --> 3110.1: for you alone, that is your iris images, your face image,

3110.1 --> 3115.3: all signed by this orb that authenticates that the data is indeed correct,

3115.8 --> 3118.1: but it belongs to you and you alone.

3118.8 --> 3122.3: And then you can process this into an iris code

3122.3 --> 3126.1: and send it to the AMPC to prove that you're a unique individual

3126.1 --> 3128.0: that hasn't signed up before.

3128.6 --> 3131.8: And then you get this additional piece of evidence of uniqueness back.

3134.3 --> 3139.2: And then maybe you have a government-issued ID card

3139.2 --> 3146.0: that is digitally signed, and you can add that to your list of evidence documents,

3146.0 --> 3149.1: so to say, that you hold privately.

3150.2 --> 3153.3: And you can say that, well, the face that signed up with this orb

3153.3 --> 3156.4: matches the face that is in this passport.

3158.1 --> 3162.5: Therefore, these documents all belong to the same individual that is me.

3163.1 --> 3166.1: And now you're starting to build this graph, right?

3166.1 --> 3168.1: The pieces are starting to connect together.

3169.1 --> 3174.5: And then, I don't know, you maybe have a signed school certificate

3174.5 --> 3178.1: that has your legal name on it that matches the name that is in the passport.

3178.5 --> 3183.9: And this is how these pieces of evidence link together

r.

3184.2 --> 3188.0: And then you can start building locally on your device

3188.0 --> 3191.9: zero-knowledge proofs over all of this and only disclose

3192.5 --> 3194.6: what it is that you want to disclose.

3194.6 --> 3202.7: Like, hey, this is a unique human being that has a bachelor's in history.

3207.5 --> 3210.7: Yeah, I guess the question I would have there is,

3210.9 --> 3218.2: what do you think the, like, at what point is enough information

3220.1 --> 3224.6: self-sovereign for you to be able to do that effectively

3226.7 --> 3229.1: is the question that comes to mind.

3229.4 --> 3231.5: Like, is it a certain number of credentials?

3231.9 --> 3234.9: Is it a certain, like, veracity or strength of the credentials?

3237.8 --> 3245.8: I think this depends largely on what it is you're trying to prove

3245.8 --> 3250.7: and what standard of proof the relying party is willing to accept.

3252.4 --> 3253.7: Yeah, makes sense.

3254.7 --> 3256.5: So, we had a question from the room.

3256.8 --> 3258.7: Kato's question is still present.

3260.2 --> 3261.6: It was more a comment.

3262.2 --> 3263.9: I was just actually typing it to you.

3264.2 --> 3268.5: So, a lot of my girlfriends, I can see this being really useful

3268.5 --> 3270.6: switching over from, like, agents.

3270.6 --> 3275.3: So, a lot of my friends, like, GPT has become recently annoying.

3275.8 --> 3278.2: And so, switching over to Gemini or Claude.

3278.7 --> 3281.3: And so, there's a lot of sunk cost fallacy of, like,

3281.3 --> 3286.5: oh, I don't want to rebuild my history with this agent.

3287.4 --> 3291.3: And so, they kind of stick to GPT or they have to build up that history.

3291.4 --> 3297.6: So, I can see this being really useful in exporting certain data from, like, a GPT

3297.6 --> 3300.3: and then importing it into a different agent.

3301.0 --> 3303.1: And I think that also the other interesting thing is

3303.1 --> 3306.9: you could also export certain data about yourself that's real

3306.9 --> 3310.1: and then add spoof data to see how the AI responds to you

3310.1 --> 3312.3: if you were, like, a different person.

3312.6 --> 3319.3: So, hi, I'm 24 years old and I'm a grad student or so

mething like that.

3319.5 --> 3321.5: Then add, like, some fake data.

3321.6 --> 3323.7: Oh, I'm actually a math genius, for example.

3324.3 --> 3327.9: So, that could be really interesting and kind of transcending

3329.9 --> 3332.6: how an AI would typically respond to you.

3333.4 --> 3334.7: Yeah, that is interesting.

3335.2 --> 3336.9: And it's funny you mentioned that

3336.9 --> 3340.3: because that was part of the motivation of me building this

3340.3 --> 3345.6: was I've been running, like, all the research through lots of different AI models

3345.6 --> 3347.3: and lots of different AI systems.

3347.3 --> 3350.6: And I needed to find a way to, like, take ownership of that

3350.6 --> 3354.6: to be able to then insert it into sort of a new model.

3355.1 --> 3361.4: So, I didn't have that, like, lock in with the specific provider.

3364.6 --> 3367.2: So, yeah, it's a good suggestion on the path.

3372.3 --> 3373.1: What time?

3373.9 --> 3374.8: We're at 2.50.

3374.8 --> 3376.2: How long have we got?

3382.5 --> 3383.2: Okay.

3383.4 --> 3385.3: So, we've got half an hour remaining.

3386.5 --> 3392.6: The person from human tech didn't arrive unless I've missed them.

3393.4 --> 3393.9: Double check.

3395.8 --> 3396.0: Nope.

3400.1 --> 3407.9: So, the 6, 7, 8 is privacy considerations and risk across all approaches,

3408.4 --> 3410.4: integration with self-sovereign identity frameworks,

3410.6 --> 3412.0: which we've kind of spoken about,

3412.6 --> 3416.5: and then the agent connection, which we're also speaking about.

3417.4 --> 3424.2: So, I'm just going to quickly brief the room on how this work is taking shape.

3425.0 --> 3428.7: So, we've been writing a document since the last block,

3428.9 --> 3431.2: called the sort of proof of personhood,

3431.2 --> 3436.8: where we're evaluating different protocols, different systems,

3437.1 --> 3439.0: different frameworks for proof of personhood.

3439.7 --> 3444.5: So, firstly, inviting everyone in the room to contribute to that paper.

3445.1 --> 3449.3: Right now, it's at the stage where we've identified what is important

3449.3 --> 3453.0: and what we're going to be following in terms of research

arch

3453.0 --> 3455.6: and some of the various examples, including WELD.
3456.4 --> 3461.1: And, yeah, we're looking for research contributions
3461.1 --> 3467.5: and also contributors to widen the amount of authors
on the paper
3470.0 --> 3472.5: over the next six months leading into the next block
meeting.
3473.8 --> 3478.1: So, I can share that paper with the group as well.
3478.7 --> 3482.2: I guess I'll also prompt the WELD guys.
3482.2 --> 3489.2: How has your experience of that process of documentat
ion through BEGIN been?
3489.8 --> 3494.6: And how would you suggest we continue?
3497.4 --> 3504.9: Yeah, Ian, as the other editor on the paper, if you'r
e available.
3506.8 --> 3513.6: Yeah, I've found that when we've worked on the docume
nt together
3513.6 --> 3521.9: pretty synchronously, it's been very productive.
3523.5 --> 3534.7: I guess I'm unclear on what exactly is most helpful f
or me to do in contributing.
3539.6 --> 3541.2: Okay, that's good feedback.
3543.6 --> 3548.3: So, when we're in the working group somewhere and we'
re in the room, we get good progress,
3548.5 --> 3551.5: but in terms of the asynchronous contribution to the
paper,
3551.7 --> 3552.7: we need a little bit more clarity.
3554.2 --> 3555.2: Is that the feedback?
3557.2 --> 3557.6: Yes.
3559.8 --> 3560.3: Cool.
3560.8 --> 3561.9: Do you have any suggestions?
3570.8 --> 3573.4: Not immediately coming to mind, but I apologize.
3573.8 --> 3579.2: I am in the U.S. and it's about 1 a.m. for me, so I a
m a little bit tired at the moment.
3579.4 --> 3581.8: I've been in your position before. I'm sorry for call
ing on you.
3583.1 --> 3589.8: Let me just - I stopped sharing the screen because I
was going to grab the paper
3590.6 --> 3591.7: and share it with everyone.
3597.4 --> 3599.3: Yeah, so this is where we're at.
3600.0 --> 3603.6: It's proof of personhood, biometric, social trust, an
d dual agent systems.
3604.2 --> 3608.1: So I can keep talking about the dual agent systems an
d why I think it's important
3608.1 --> 3618.6: to have this clear separation between cryptographic k
eys and agency or knowledge.
3624.7 --> 3629.8: I'm just going to scroll through so you guys can obse
rve what it's looking like so far.
3630.9 --> 3642.0: So the clear problem statement is very much to do wit

h AI, with the flavor of blockchain.

3644.0 --> 3647.4: We've got biometric-based systems, social proof systems.

3648.4 --> 3653.2: We speak about the end users. Privacy is important. Privacy is normal.

3666.4 --> 3668.5: Let me get into agent duality.

3671.3 --> 3673.7: And here are the contributing organizations.

3673.7 --> 3682.3: So if there are any organizations that are not named here, I would like feedback in that regard.

3686.2 --> 3691.9: Yeah, let me know or reach out directly if you'd like to sort of name a new organization for us to work with.

3693.3 --> 3699.6: Many of these I've already got connections with, but if there are new ones, also let me know.

3701.5 --> 3705.8: And this, did you have a question?

3713.7 --> 3719.5: So process-wise, to progress a document, I have one suggestion.

3721.1 --> 3733.8: Maybe we should just send the document out to the contributors with a set deadline for comments.

3735.2 --> 3742.9: And we set an agenda for the certain date meeting.

3742.9 --> 3754.8: And by that time, if we get enough buy-in or enough pushback, we decide to accept the comments or contribution,

3755.2 --> 3759.2: or decide to modify it or something like that.

3761.6 --> 3765.7: The document is, a lot of sections are still quite empty.

3766.2 --> 3771.0: But once the document is filled up appropriately,

3771.0 --> 3780.2: perhaps we could also try to communicate to the somewhat wider community as initial draft or something like that

3780.2 --> 3784.3: to solicit more comments on it, to improve the document.

3784.8 --> 3800.4: So that we can target the next block to make it a fast release version or something like that.

3801.0 --> 3810.7: Because we are in such a quickly shifting environment, and we can't take two years to complete the document.

3810.9 --> 3814.8: Like in ISO, it's two, three, four years, right?

3815.5 --> 3820.5: But by the time four years pass, I think it's going to be pretty stable.

3821.1 --> 3828.1: So we should probably try to have some kind of...

3828.1 --> 3829.5: Like a deadline schedule.

3829.5 --> 3832.6: Okay, that's a good suggestion. Thank you.

3833.6 --> 3844.2: So we'll make sure that the hello.ai transcript, make sure that that is included.

3851.6 --> 3859.4: Documents. A document is going to be only as good as the contributions, right?

3859.8 --> 3861.5: It's not going to be any better than that.

3864.0 --> 3871.0: So your cooperation and your contribution is really important.

3871.0 --> 3874.7: And I'm looking forward to have your contributions to

this document.

3884.9 --> 3888.8: Just a quick question. I'm curious how ISO standardization works.

3889.0 --> 3895.1: And if this is ultimately, if this is the foundation of the startup of that kind of process, right?

3895.1 --> 3896.8: Getting stakeholders from the industry.

3897.4 --> 3904.9: And then ultimately, does this expand to stakeholders and the nation state level to contribute to a document?

3905.2 --> 3913.4: So if you're looking at standardization processes that multiple years of this stage or, you know, 12 to 18 months of this stage.

3913.5 --> 3915.2: And then where does it progress?

3915.8 --> 3920.5: I know a few folks in the room are very involved in ISO standards bodies.

3923.3 --> 3930.0: So not as a co-chair of this working group or anything to do with BGIN, really, my personal opinion.

3930.6 --> 3938.2: But BGIN, first of all, is a class A liaison to TC307 in ISO, right?

3938.5 --> 3942.5: That means we can actually, we may also pass submitter.

3942.7 --> 3944.8: Submitter? I don't know.

3945.5 --> 3951.7: That means we can actually submit a document to ISO for publication.

3952.2 --> 3959.8: Now, this kind of document clearly isn't a standard because it doesn't have any should or shall, right?

3960.5 --> 3962.3: It's just reporting out.

3962.3 --> 3966.0: So there are three types of document in ISO.

3967.0 --> 3969.0: IS, international standard.

3969.6 --> 3971.5: TS, technical specification.

3972.1 --> 3974.6: And TL, technical report.

3975.1 --> 3980.4: And this kind of document belongs to the TR category because it's a survey, right?

3982.5 --> 3986.9: The survey and summary of the state of the world.

3986.9 --> 3996.0: So potentially, once this document is completed to a reasonable state,

3996.4 --> 4007.5: then we could submit to TC307 for publication as TR, technical report.

4008.4 --> 4011.5: And technical report takes only one voting.

4013.0 --> 4014.3: Four months.

4015.2 --> 4017.6: So that's much shorter, right?

4018.3 --> 4023.5: Well, in the case of normal IS, international standard,

4023.9 --> 4026.8: you have to start from preliminary work item.

4027.0 --> 4029.3: Then it goes to approved work item.

4029.6 --> 4036.0: You have to develop working draft, committee draft, draft international standard,

4036.8 --> 4039.9: final draft international standard, and publication.

4039.9 --> 4045.5: And each takes more than six months, typically.

4048.7 --> 4055.5: Building on that, I'd be curious if the solutions that are developed in rooms like this

4055.5 --> 4060.5: would take precedence and importance and ultimately end up becoming the standard

4060.5 --> 4064.6: if perhaps in parallel you have your Google or Apple of the world

4064.6 --> 4068.5: that will at some point develop their own proof of personhood solution

4068.5 --> 4074.3: to the extent that they need to solve a problem that their users are experiencing.

4074.8 --> 4082.6: If it's like a first to develop the standards or first to solve a commercial problem

4082.6 --> 4089.6: and obtain a critical mass of users that are using an other proof of person solution

4089.6 --> 4091.5: that's already out there, right?

4091.5 --> 4097.0: How does an ISO interest group face that?

4097.0 --> 4105.0: Is that something that's pretty typical in a standards process or in the industry?

4116.0 --> 4118.1: Do you want to speak?

4121.0 --> 4122.8: So do you want the reality or the theory?

4124.0 --> 4125.2: Let's go with both.

4125.2 --> 4131.0: So the reality is if Apple and Google decide to do something, it doesn't matter what we say.

4132.1 --> 4135.0: You know, that is the de facto standard.

4135.4 --> 4137.3: We can do a de jure standard.

4137.3 --> 4142.5: If it works well enough, then we could actually probably get Apple and Google to sign on

4142.5 --> 4144.8: and then that really works.

4145.6 --> 4150.7: The nightmare scenario is we come up with the theoretically best solution

4150.7 --> 4155.4: and they come up with crap, the world's going to have crap

4158.0 --> 4160.1: from 30 years of doing standards.

4165.4 --> 4168.6: Directionally, it sounds like it flows in one direction, right?

4168.9 --> 4174.6: Have there been interesting use cases for standards that have flowed the other way

4174.6 --> 4177.0: where the bodies have come up with a standard?

4177.0 --> 4181.1: It's then been incorporated and adopted by the big players?

4184.8 --> 4189.2: Honestly, and this is how the ecosystem kind of works.

4189.8 --> 4194.4: People like us will come up with great ideas, get it at least half-baked

4194.4 --> 4200.7: and bring it to ISO or IETF, whatever makes sense.

4203.6 --> 4207.4: The standards that never get used is when you go to the standard body

4207.4 --> 4209.2: and say, we want a personhood thing.
4209.3 --> 4210.3: Let's talk about it.
4210.3 --> 4211.7: Let's figure out how to do it.
4212.1 --> 4218.2: And then you just spend years with professional standards personnel
4221.4 --> 4225.0: who don't have product responsibility at their companies.
4225.5 --> 4227.7: And then you might even come up with something good,
4227.9 --> 4229.7: but then the company is like, yeah, why do we care?
4229.7 --> 4233.5: As opposed to hammering out things that people,
4233.9 --> 4237.2: like you've got something that you can say,
4237.3 --> 4241.9: here's our real world experience with moving the data around,
4242.1 --> 4246.2: charting the data, doing proofs.
4246.4 --> 4248.2: Because the other thing you find is people go,
4248.3 --> 4250.4: oh, you can't possibly make that work.
4251.6 --> 4255.9: And so when you say, right, it's kind of like video and voice
4255.9 --> 4259.3: over a packet network, it can't possibly work.
4260.6 --> 4262.4: You've got to have guaranteed quality service.
4262.6 --> 4266.9: You can't possibly have voice and video like we have right now.
4276.8 --> 4280.8: Could you talk more about the dual agent systems aspect of this?
4280.9 --> 4284.4: Because I've always been interested in the proof of personhood aspect.
4284.9 --> 4287.8: We're kind of trying to prove that we're humans and not bots.
4287.9 --> 4292.1: But I think the natural evolution of what we've seen already is that
4292.1 --> 4296.5: we can definitely prove that we are people,
4296.7 --> 4299.7: but we're not necessarily proving that the content that we're providing
4299.7 --> 4303.2: is our own output and it's not just kind of AI output.
4303.8 --> 4307.7: But the natural trajectory of all this is obviously what we would be working
4307.7 --> 4310.4: with our own agents and those agents might be talking to each other
4310.4 --> 4311.5: and we would be inputting.
4311.5 --> 4315.2: But I'm just curious from the context of proof of personhood,
4315.8 --> 4319.7: what's the dual agent angle there?
4323.2 --> 4324.9: Yeah, happy to share that.
4325.1 --> 4327.5: And that's to do with my research.
4328.0 --> 4332.4: One sec, let me just grab a diagram that helps explain this.
4335.4 --> 4336.8: Yeah, here it is.

4337.8 --> 4342.0: So this is a diagram that expresses the human stack
4342.9 --> 4347.2: and it mirrors the IP.
4347.8 --> 4352.0: So the IP stack, which all of the machines work on,
4352.2 --> 4356.3: has IP as the spanning protocol in the middle.
4356.9 --> 4358.3: And at the top, you've got software.
4358.7 --> 4360.2: And at the bottom, you've got hardware.
4361.1 --> 4363.0: This is the human stack.
4363.0 --> 4368.2: And the human stack has the UI and the trust tasks at
the top
4368.2 --> 4371.9: and the credentials and all of the magic at the botto
m.
4373.6 --> 4379.5: Dual agent architecture and why you need separation a
nd duality here
4379.5 --> 4386.0: is because these delegate and protect conflicting con
straints.
4386.5 --> 4388.5: So when you think about information,
4388.5 --> 4395.0: if I was to want to establish a friendship or a crede
ntial with you,
4395.6 --> 4396.7: I've got a choice.
4397.3 --> 4401.4: I can either block you or I can let more information
through.
4401.9 --> 4404.2: And if we choose to let information through,
4404.4 --> 4407.4: then we are establishing understanding between each o
ther
4407.4 --> 4410.9: and eventually we will reveal more information to eac
h other
4410.9 --> 4412.1: because we trust each other more.
4412.4 --> 4415.7: So maybe on our first call, we don't have video on,
4415.9 --> 4417.3: but on the second call, we have video.
4417.3 --> 4418.7: So we trust each other.
4419.0 --> 4421.6: That has gone through a trust task.
4423.3 --> 4428.7: Now, if those conflicting constraints are within one
system,
4428.8 --> 4434.2: so one agent has to deal with protecting information
from someone
4434.2 --> 4440.1: and the same agent has to deal with sending and compo
sing information to share,
4441.1 --> 4446.0: it will collapse into one surface and not actually be
able to,
4446.1 --> 4448.7: like you can, it will be easily attacked
4448.7 --> 4454.5: because it has this like internal conflict of it does
n't actually know
4455.1 --> 4456.9: whether or not this information should be shared
4456.9 --> 4458.7: or this information should be secure
4458.7 --> 4462.4: because each sort of interaction is different.
4463.1 --> 4464.0: So what do you do?
4464.6 --> 4468.2: You split the system into two agents.

4468.2 --> 4472.2: One agent deals with the boundaries,
4472.8 --> 4475.9: so boundary making and protecting information,
4476.1 --> 4478.2: and the other agent composes information
4478.7 --> 4480.6: and has access to the knowledge substrate.
4481.2 --> 4486.0: So the knowledge substrate, that like pretty picture
of the spell web,
4486.5 --> 4489.4: that's the top, and I call that the mage,
4489.9 --> 4494.1: and your keys, so biometric proof from world,
4494.1 --> 4499.1: would be one of these VC slash crypto keys at the bot
tom.
4500.0 --> 4504.5: And that wallet should never touch knowledge substrat
e.
4505.2 --> 4511.5: And the only way that that's possible is you create t
wo separate systems
4511.5 --> 4514.4: that never have that overlap of information
4514.4 --> 4520.0: and you try and prove that you've mathematically sepa
rated those systems
4520.0 --> 4525.3: into two using trusted execution environments or some
thing like that.
4526.0 --> 4530.3: And I think that this is going to be the natural stat
e long term
4530.3 --> 4536.0: for using AI of having these conflicting constraints
4536.7 --> 4539.3: as the duality, as the base duality,
4539.8 --> 4542.7: and then obviously you can expand out from there.
4543.7 --> 4547.6: But that's what the dual agent protocol and thesis is
about.
4547.6 --> 4551.5: And what I was speaking about on the graph
4551.5 --> 4557.4: and why the graph has those two connect and reflect q
ualities
4557.4 --> 4564.7: is they're the emergent properties of the agents
4564.7 --> 4568.1: creating their own secret language between each other
4568.6 --> 4570.5: because they can't share information.
4571.3 --> 4572.5: So they need to communicate,
4572.7 --> 4574.9: they need this like trust spanning protocol in the mi
ddle,
4574.9 --> 4582.7: and the emergent quality of the top is to be able to
reflect on information.
4584.0 --> 4593.1: And the emergent quality of the top is to be able to
connect information.
4594.9 --> 4598.9: So now I'm happy to connect knowledge with you
4598.9 --> 4600.4: because we've performed some trust tasks.
4600.4 --> 4604.7: And the emergent quality of the bottom is to be able
to reflect.
4605.4 --> 4609.7: And how you reflect is you add time to the equation.
4610.4 --> 4614.6: And in my opinion, the only way to properly add time
is using blockchain.
4615.4 --> 4618.4: But obviously you can use other systems.

4618.9 --> 4621.6: But if I make a discrete measurement on a blockchain
4621.6 --> 4626.5: at a particular point in time and that agent made that measurement
4626.5 --> 4631.5: and the one above, the major agent that's observing that,
4632.2 --> 4637.2: is like, yep, that was based on the composition of data and information
4637.2 --> 4639.9: that I then put in a zero-knowledge proof
4639.9 --> 4643.7: and shared to be true with the keys.
4645.4 --> 4648.0: Then you have another event in the future.
4649.2 --> 4652.5: Over time, you get this reflect emergent quality
4652.5 --> 4656.9: where the agents don't share information,
4657.1 --> 4660.6: but they're able to infer information based on the gap between them.
4661.4 --> 4664.4: And it's that gap between them that I'm most interested in
4664.4 --> 4669.1: because that's how you create agent communication,
4669.6 --> 4673.0: which has polarity and like a plus and minus.
4673.6 --> 4678.6: So one agent gives and the other one receives information.
4679.8 --> 4683.1: And, yeah, duality is important.
4685.6 --> 4688.5: So that's where the research direction is going.
4689.2 --> 4692.2: And it's expanded in a whole bunch of different ways.
4692.6 --> 4697.9: This is the overlap with the layer of the agent systems,
4698.2 --> 4703.4: but there's a few other documents that I can share as well.
4704.4 --> 4707.1: And here's the document suite of the living.
4707.7 --> 4712.6: So, yeah, it expands out.
4713.5 --> 4715.4: One thing that I would like to mention
4715.4 --> 4718.9: which came to mind around the standards process
4719.4 --> 4722.9: was one of the standards that I've been involved in recently is my terms,
4723.1 --> 4726.9: the IEEE 7012.
4727.5 --> 4729.7: I think this may actually be a case
4729.7 --> 4733.0: where they spent eight years developing the standard
4733.7 --> 4737.7: and now it's finally happening and hopefully it gets adopted.
4739.0 --> 4740.9: And right now it's the process of adoption.
4741.9 --> 4743.2: So I don't know.
4743.4 --> 4745.7: Maybe right now I'm just thinking,
4746.6 --> 4748.9: is there some advice on that path
4748.9 --> 4752.5: where it came from the standards organization,
4752.7 --> 4755.8: the idea was the standard, it was important,
4756.0 --> 4757.5: and let's talk about it.
4757.5 --> 4760.8: And now eight years later it's finally become a standard,

4761.5 --> 4764.9: but it didn't come from a place of people building
4765.5 --> 4766.7: and then informing it.
4766.8 --> 4768.1: We now need to build it.
4768.5 --> 4770.2: So any advice on that?
4772.0 --> 4775.1: So there are lots and lots of standards,
4775.5 --> 4779.3: whether ISO or ITF, IEC or IEEE, it doesn't matter.
4780.5 --> 4783.0: Gets published, never used,
4784.0 --> 4789.3: apart from a couple vendors who are proposing that.
4801.3 --> 4804.3: Because I'm the author of that specification,
4804.8 --> 4808.4: I can use that as a counterexample,
4808.6 --> 4811.0: but OpenID Connect, like World is also using,
4811.0 --> 4821.3: is being adopted by so many companies and countries
4821.3 --> 4825.4: long before it got published as IIS.
4836.1 --> 4839.1: Now that all said, and I'm trying to think,
4839.1 --> 4841.3: but I know that I've got examples
4842.1 --> 4844.5: where something like this happened, it took forever,
4845.2 --> 4847.7: and that's part of, that's a big red flag.
4848.3 --> 4851.7: Because if it took eight years, nobody really cared.
4853.1 --> 4856.3: Because if somebody cares, you get something
4856.3 --> 4861.1: that almost works out and then deal with my terms too.
4864.4 --> 4867.7: But sometimes something like this gets out there
4868.9 --> 4871.3: and actually part of it was nobody cared,
4871.4 --> 4872.8: but it was also ahead of its time.
4873.6 --> 4878.7: And now the time has come, way better to pick this up.
4879.3 --> 4882.1: And if needed, my terms 2.0,
4883.2 --> 4886.6: then to say, oh, let's come up with a standard
4886.6 --> 4889.6: for machine readable personal privacy,
4889.8 --> 4892.0: and then we take eight years because nobody cares.
4892.5 --> 4895.4: And 16 years from the beginning of this,
4895.4 --> 4896.5: we end up with the same thing.
4899.8 --> 4902.1: No need to make the same mistake someone else made.
4903.3 --> 4904.5: Yeah, reinvent the wheel.
4906.4 --> 4909.0: So I'm in the opinion that there has to be
4909.0 --> 4915.7: some kind of smooth entry point to the spec family.
4915.7 --> 4916.9: All right.
4918.6 --> 4926.3: If the hurdle to get the first bump hump over is too
hard,
4926.9 --> 4928.9: it's really a hard sell, right?
4929.2 --> 4930.9: So in the case of OpenEdge Connect,
4931.7 --> 4934.4: the easiest part is really easy.
4934.7 --> 4935.6: If you're running a party,
4936.4 --> 4939.4: and if you just wanted a basic profile,
4939.6 --> 4941.9: you could probably code it in one afternoon.
4942.9 --> 4945.7: And there's a more esoteric part,
4946.6 --> 4949.0: who a lot of people said,
4949.2 --> 4952.4: it's not going to be ever used, right?
4953.7 --> 4956.0: They proved to be wrong.

4956.1 --> 4959.7: After like nine years, all the banks are implementing it.

4962.9 --> 4968.5: But then if it was only that, that hard part,

4968.5 --> 4972.6: nobody probably looked at OpenEdge Connect to start with.

4973.0 --> 4976.7: So the second part wouldn't have happened either.

4979.6 --> 4985.7: So everybody can, unless you have some leg problems,

4985.8 --> 4986.4: something like that,

4986.4 --> 4994.3: you probably can climb up to the second floor with a stair, right?

4994.8 --> 4996.4: But if there isn't any stair,

4997.4 --> 5000.7: it's really hard for people to climb to the second floor.

5001.0 --> 5002.3: That's a good analogy.

5015.5 --> 5018.2: Do you know why, in this case,

5018.3 --> 5022.7: they decided to go the standards route first

5022.7 --> 5026.2: before just building it?

5028.8 --> 5030.3: It's a good question, actually.

5032.4 --> 5035.1: Is it more to do with the background of the people who were involved?

5035.2 --> 5038.0: I think it's to do with, so it came out of Project VRM,

5038.8 --> 5041.3: which is Vendor Relationship Management.

5043.2 --> 5051.4: And I think DocSales is sort of the lead behind this standard.

5051.4 --> 5054.0: And he wrote the intent economy.

5054.9 --> 5057.8: And that's why the comment from Eric is important.

5058.2 --> 5063.5: This may actually be the time has come for this

5063.5 --> 5066.0: because AI is consuming all of our data

5067.1 --> 5072.5: in a way that we don't have control or ownership of it.

5074.5 --> 5077.3: So I think that they went down this path

5078.3 --> 5082.1: because they wanted, when time did come,

5082.3 --> 5086.4: to have the oomph of it being a standard

5086.4 --> 5089.6: so then people felt comfortable adopting it.

5090.4 --> 5093.9: I also think they observed GDPR

5095.3 --> 5100.2: and sort of saw both the benefit that that created

5100.2 --> 5107.2: but also some of the new incumbents that were created from that.

5107.5 --> 5109.5: There are maybe four or five providers

5109.5 --> 5111.7: that do most of the cookie banners in the world

5111.7 --> 5114.4: and they've now become their own data oligopoly.

5116.7 --> 5123.0: So I think that they wanted to make it a standard

5123.0 --> 5125.8: rather than just doing it themselves

5125.8 --> 5129.1: and having this sort of risk of centralization.

5129.1 --> 5131.6: So it was more permissionless for people

5131.6 --> 5135.1: to just adopt their own machine-readable personal pri

vacy terms.

5140.8 --> 5143.3: Yeah, this all links together
5144.1 --> 5146.5: because hopefully a personhood credential
5147.1 --> 5150.3: gives you the ability to create these relationship agreements
5151.1 --> 5154.9: with the websites and the different systems that you visit
5154.9 --> 5158.1: which then creates that sort of more bilateral
5158.9 --> 5162.8: data contribution agreements to become available.
5174.5 --> 5177.2: So we have 10 more minutes.
5180.5 --> 5183.2: However, I'm going to open it to the room
5183.2 --> 5185.6: if there are any other comments or questions.
5187.2 --> 5189.3: But if there are not, perhaps we can.
5191.3 --> 5192.3: The right ending closes.
5195.1 --> 5197.4: Oh, is it 20 past not half past?
5198.6 --> 5199.8: We have one minute.
5201.5 --> 5204.0: So in terms of next steps,
5204.2 --> 5208.6: we'll continue this work in the working group
5208.6 --> 5210.3: and take on the feedback.
5210.3 --> 5212.6: A meeting report of this meeting
5212.6 --> 5214.9: is going to be distributed to all of the participants.
5215.8 --> 5219.3: So feel free to sort of observe that and provide feedback as well.
5220.3 --> 5224.4: But yeah, I think hopefully the goal
5224.4 --> 5227.3: is to have the personhood paper at a stage
5227.8 --> 5234.1: where we can go down the TR path of the ISO path
5234.1 --> 5240.2: and present it in a coherent way
5240.2 --> 5242.6: that opens it up to the greater public for comment
5242.6 --> 5247.0: in block 15 in October in DC.
5247.4 --> 5250.3: So thank you all for participating in this working group
5250.3 --> 5252.0: on personhood today.
5275.6 --> 5278.8: And the next session will start at 3.30.
5279.0 --> 5282.8: And it will be on stable coins in this room.
5322.7 --> 5324.1: Thank you.
5352.7 --> 5354.1: Thank you.
5358.4 --> 5359.8: Thank you.